

Research Candidate Mini Project

Predicting Human Survey Response Distributions with SimBench

Deadline: 2 weeks from assignment date

Overview

This case study addresses the following problem: given a survey question and its answer choices, predict the distribution of responses a real human population would give. We use the [SimBench dataset](#) as ground truth (read the paper [here](#)). The objective is a principled, LLM-based method that improves on the naïve baseline reported in the SimBench paper and generalizes to questions and population segments not seen during development.

The case study has two parts. **Part I** (Simulation Implementation) is a hands-on exercise: build the prediction method and evaluate it. **Part II** (Feedback Mechanism Architecture) is a systems-design exercise, with no implementation required: assuming a commercial deployment of the simulator, propose an architecture for feeding real-world outcomes back into the model so that its predictions improve over time.

The task is intentionally open-ended; it is meant to test both creativity and rigor, with weight on formal justification, sound evaluation, and clear reasoning. The broader question behind it is whether large language models can serve as reliable simulators of population-level opinion across varying demographic conditions.

Part I — Simulation Implementation

Task Definition

The task is to predict the empirical distribution of human responses recorded in the SimBench dataset, given:

- a survey question,
- a discrete set of answer choices,
- an optional conditioning segment of the population (e.g., age range, gender).

The conditioning segment may be empty, corresponding to the full population, or specified in text. The method must generalize to questions and segments not seen during development, and must not rely on access to ground-truth answers.

Materials and Method

The core modeling mechanism must be LLM-based; beyond that, the methodology is open. You may use any experimental design, inference strategy, or auxiliary data source, provided it does not leak ground-truth information or otherwise exploit dataset artifacts to “cheat”.

Permissible extensions include, but are not limited to:

- persona-based conditioning (e.g., synthetic personas derived from external datasets such as [Nemotron](#)),
- leveraging data from other datasets or the SimBench dataset itself (while avoiding ground-truth leakage),
- alternative elicitation or prompting formats for LLM responses,

- the use of tools, including ones that access the internet to gather information.

Validate every extension with an ablation that isolates its individual contribution to performance.

Evaluation Metrics

Evaluate performance with the following metrics.

SimBench Score

The primary metric is the SimBench score S , as defined in the paper: a measure of similarity between the predicted answer distribution and the empirical human response distribution.

Counterfactual Sensitivity Score

Alongside overall distributional accuracy, include a counterfactual sensitivity score. This measures whether conditioning on a specific subgroup (e.g., a demographic slice) shifts the predicted distribution in the correct direction relative to the full population.

Formally, it compares:

- the predicted difference between the conditioned and unconditioned distributions, and
- the corresponding ground-truth difference observed in the data.

The aim is to test whether the model updates its beliefs in the right direction when conditioning information changes, independent of how accurate the unconditioned prior is. Design the metric as you see fit to capture this.

Experimental Scope

Your evaluation must include, at a minimum, predictions for the following SimBench questions:

- “Would you say you have a lot, some, a little or no trust in the president?”
- “Gay men and lesbians should be free to live their own life as they wish.”
- “How often do you use the internet on devices such as computers, tablets, or smartphones, whether for work or personal use?”

You may include additional questions at your discretion.

Assignment Objectives

Part I is evaluated along the following dimensions:

1. **Modeling and inference quality.** Does the method predict empirical response distributions accurately, and does it meaningfully improve on the baseline?
2. **Use of conditioning information.** Does it incorporate demographic or subgroup conditioning correctly, and update predictions in the right direction?
3. **Technical justification.** Is the method formally motivated, with assumptions and limitations clearly stated?
4. **Experimental rigor.** Are the evaluation metrics well-designed, and do the experiments (including ablations) give confidence in generalization?

Part II — Feedback Mechanism Architecture

Part I treats the simulator as a research benchmark; Part II asks how it behaves as a commercial product. This is a systems-design question, not a coding task: the deliverable is an architecture

diagram and the reasoning behind it, with no implementation required. The scenario below represents the central requirement for converting incoming real-world data into a continuously improving model for commercial deployments.

Suppose we are running a customer-population simulation for a sports e-commerce partner. Until now the simulator has predicted population-level behavior — purchase propensity, response to offers, differences across segments — with no real-world outcome data for the predicted results. The engagement now begins to return real outcomes:

- **Revealed behavior:** transaction logs for real individuals — what they bought, when, and in response to what.
- **Stated responses:** new survey waves fielded to the same or an overlapping population.

As this data arrives, the simulator must improve. Propose an architecture for feeding these real-world outcomes back into the system. At a minimum, address:

1. **The compounding memory.** What is stored, and how does it feed predictions?
2. **Method.** Lay out the spectrum — from in-context retrieval through to supervised fine-tuning — and justify what you would use where.
3. **Signal types.** How do you treat revealed behavior (purchases) differently from stated preference (surveys)?
4. **Generalization and multi-tenancy.** Does ground truth from one engagement improve predictions for others?
5. **Guardrails and evaluation.** How do you guard against overfitting and drift, and how do you validate an update before it ships?

Infrastructure and System Setup

As part of the presentation, please briefly address the practical infrastructure you would use to run this project on a larger scale. In particular, discuss:

- What compute setup you would choose (e.g., local machines, cloud instances, reserved or on-demand GPUs), and why.
- How you would access and run language models (e.g., hosted APIs, inference servers, model hubs).
- How data would be stored and accessed during experimentation.
- How you would manage the process of running experiments on different changes and tracking their results.

There is no single correct setup. The goal is to assess clarity of reasoning, feasibility, and awareness of tradeoffs rather than specific tooling choices.

Performance Criteria

Submissions are assessed according to the following criteria:

1. **Creativity and novelty.** Does the approach introduce new ideas or meaningfully extend existing methods?
2. **Technical depth.** Is the problem formalized, and is the solution mathematically or algorithmically justified?

3. **Empirical validation.** Do experiments demonstrate clear improvements, supported by ablations and appropriate metrics?
4. **Implementation quality.** Is the codebase readable, modular, and easily extensible?
5. **Systems reasoning.** Are the architectural tradeoffs sound, and is the design of the feedback loop clear and well-justified?

Presentation

In addition to the written submission, prepare a presentation suitable for a 60–90 minute discussion. The presentation should clearly communicate:

- The motivation and high-level intuition behind the approach,
- Key modeling decisions and formal justifications,
- Experimental results and ablation findings,
- Limitations, risks, and potential extensions.

Deliverables

The final submission should include:

- Predicted answer distributions for the required SimBench questions,
- Quantitative evaluation and ablation using the specified metrics,
- An architecture diagram of the closed loop,
- A presentation summarizing the approach and results.